

FINAL PROJECT PROPOSAL

Evidence-Aware Validation of AI-Generated Drug Candidates Using PPI Networks and Network Pharmacology

Team Members

Dongwan Kim · Zander Xavier Rothering · Edurne Gonzalez de la Vega Arteaga
DaQuawn Donnell Edwards · Shivani Tijare · Girish Krishna

Problem

A fundamental gap exists between what generative AI models produce and what is known about molecular safety. Models such as MolGPT, REINVENT, and diffusion-based approaches are highly effective at designing compounds with high binding affinity and drug-likeness. However, they optimize against molecular property functions, not published evidence. Off-target interactions and scaffold-associated toxicities documented in the literature remain invisible to these models, meaning a structurally promising candidate may carry well-characterized risks that were never checked.

Proposed Solution

We address this gap with a post generation validation pipeline that accepts an AI generated compound (as a SMILES string) alongside its intended target protein, constructs a target centric protein–protein interaction (PPI) network from public databases (STRING, BioGRID), retrieves safety relevant literature via PubMed, and delivers a structured risk assessment before any candidate advances to synthesis.

Pipeline Overview

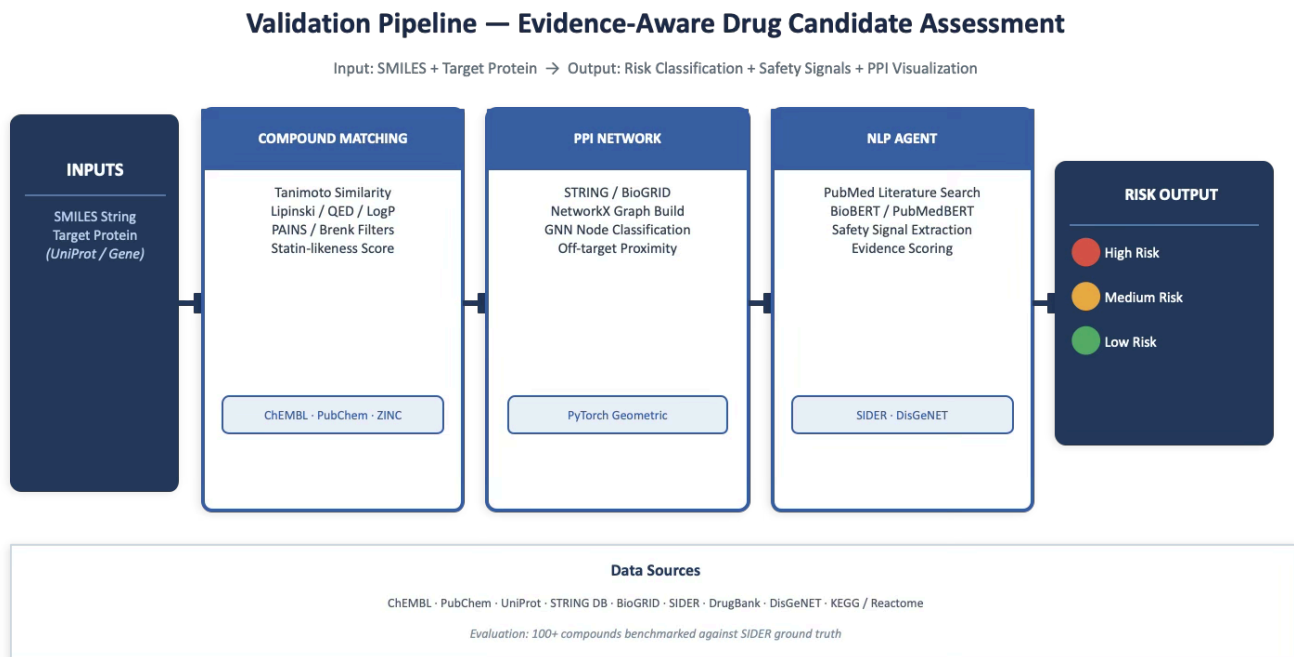


Figure 1. End-to-end validation pipeline: SMILES + Target Protein → Risk Classification

Inputs & Outputs

Inputs

- SMILES string (from a generative model or ChEMBL)
- Target protein — gene symbol or UniProt ID
- Indication / therapeutic area (optional)

Outputs

- **High / Medium / Low risk classification** with a confidence rationale
- Identified safety signals with supporting literature sources
- PPI network visualization highlighting risk-associated interactions

Technical Approach

The pipeline integrates four components in sequence, following a classical-before-neural development order:

- PPI network analysis using NetworkX and PyTorch Geometric for graph neural network (GNN) node classification within STRING/BioGRID networks, inferring biological risk through topological proximity to known off-target proteins.
- Compound matching via Tanimoto similarity, drug-likeness metrics (QED, LogP, Lipinski's Rule of Five), PAINS/Brenk structural filters, and statin-likeness scoring against known HMGCR inhibitors.
- Literature retrieval via the PubMed API, with NLP-based safety signal extraction powered by BioBERT and PubMedBERT.
- Risk aggregation that combines structural, network-topological, and literature-derived evidence signals into a single High / Medium / Low classification with traceable source citations.

The pipeline will be evaluated against 100+ compounds using SIDER as ground truth for known adverse effects; AI-generated SMILES will be sourced from the MOSES benchmark, GuacaMol, and ZINC250K datasets.